



STACK · BY STACKFORGE LABS

LAUNCH STACK.

*From the laptop to the Play Store,
one step at a time.*

1
Accounts

2
Fix the app

3
Server live

4
Release build

5
Play Console

6
12 testers × 14
days

7
Launch & grow



THE ROUTE

About **6 weeks** from today to a public Play Store launch. The slowest part is the 14-day closed test, so everything is ordered to **start that clock as early as possible**.

STAGE	WHAT	DONE WHEN	WHEN
1	Accounts & name	Play Console, domain, Anthropic key and Vercel ready	Week 1
2	Make the app store-ready	No blocked permissions, Spotify hidden, AI on Haiku	Week 1
3	Web + server live	stackforge.in serves the app, AI, privacy and delete pages	Week 1–2
4	Signed release build	An .aab signed with your own upload key	Week 2
5	Play Console setup	Store listing and all "App content" forms green	Week 2
6	Closed test	12+ testers opted in for 14 days in a row; production approved	Weeks 2–5
7	Launch & first 30 days	Staged rollout to 100%; weekly numbers	Week 6 →

HEALTH CHECK OF YOUR REPO

CHECKED IN ~/DESKTOP/PROJECTS/STACK ON 29.09.2026

CHECK	STATUS	NOTE
Target SDK	OK · 36	Meets Play's current target level (Android 16)
Secrets in git history	OK · none	.env* and google-services.json are gitignored; no API keys in any commit
GitHub remote	OK	github.com/chhari07/stackforge already matches the new name
Release signing	Missing	build-apk.sh only makes a debug APK. Play needs a signed .aab (Stage 4)
All files access	Blocker	MANAGE_EXTERNAL_STORAGE is in the manifest. Remove it from the Play build (Stage 2)
AI server URL	Emulator only	NEXT_PUBLIC_AI_URL is http://10.0.2.2:3000. Needs your https domain
Version	1 / "1.0"	Bump versionCode on every upload
Uncommitted changes	5 files	Commit before building a release so each build matches a commit



ACCOUNTS & NAME

Do these first. Some take days to be approved, so start them on day one and code while you wait.

1

Check the StackForge name

Search "StackForge" on IP India (classes 9 and 42). Then check `stackforge.in`, `stackforge labs.com` and `stackforge.app` at a registrar. Buy the `.in` and one global domain.

2

Create the Google Play Console account

`play.google.com/console` · one-time fee **US\$25**. Choose **Personal** now (quick). An **Organisation** account needs a registered company and a D-U-N-S number, but it skips the 12-tester rule. Upload your ID; Google may also ask you to confirm you have a real Android phone (Play Console app). Approval can take a few days.

Developer name on the store: "StackForge Labs".

3

Create a support email

For example `hello@stackforge.in`, using your domain's email forwarding or Zoho Mail's free plan. Play shows it publicly, so keep your personal Gmail private.

4

Anthropic API key with a spending limit

`console.anthropic.com` → create a key for "stack-prod" → set a **monthly spend limit** (for example \$20) so a bug can't run up a bill.

5

Vercel account

Sign in with GitHub and import `chhari07/stackforge`. The free Hobby plan is enough to start. It will host the web app, the AI route and the privacy pages.

6

Claim the handles

GitHub org `stackforge-labs` (optional), plus X and Instagram `@stackforge labs`. Use the same spelling everywhere.

7

Line up 15–20 testers now

Friends, classmates, dev groups. You need **12 who stay opted in for 14 days**, so invite extras. Collect their Gmail addresses in a sheet. (Stage 6 has the details.)

COST SO FAR

Play Console \$25 (≈ ₹2,200) · domains ≈ ₹1,500 a year · Vercel and Firebase free tiers · AI pay-as-you-go with a cap. **Total to launch: under ₹5,000.**



MAKE IT STORE-READY

Code changes that stop Play from rejecting the app and keep AI costs safe. Commit after each one.

1

Two builds: Play and sideload

Add a build switch (for example `STACK_STORE=play`). In the Play build, `scripts/android-setup.mjs` **leaves out** `MANAGE_EXTERNAL_STORAGE` and the "scan the whole phone" button. PDFs come in through the folder picker (SAF) and Share to Stack. The sideload APK can keep full scanning.

2

Hide Spotify

Spotify caps you at 25 users. Remove it from the Play build, or keep it under Settings → "Advanced: use your own Spotify Client ID". Phone music stays.

3

News: summary + link

For news cards, show the headline, a short summary and "Open original". Keep full reader mode only for articles the user **saved**.

4

AI: cheap by default, capped per user

Set `AI_MODEL=claude-haiku-4-5` for everyday tasks. Keep `AI_DAILY_LIMIT` low (for example 20 per user per day). The in-memory limit resets when Vercel restarts, so move the counter into Firestore before you have many users.

5

Point the app at the real server

The APK build uses `NEXT_PUBLIC_AI_URL=https://stackforge.in` (or the Vercel URL) instead of `10.0.2.2`. The emulator cleartext rule should apply to debug builds only.

6

Account deletion inside the app

Play requires apps with sign-in to let users **delete their account and data**. Add Settings → Account → "Delete account": remove `users/<uid>` from Firestore, the user's Storage files and the Auth user.

7

Crash and usage numbers

Add something simple to count installs, active days and reviews done (Firebase Analytics or your own Firestore counter). You'll need day-1, day-7 and day-30 numbers.

8

Version numbers

In `android/app/build.gradle`, set `versionCode 2` and `versionName "1.0.0"`, and add 1 to `versionCode` for every upload.



SERVER & WEB LIVE

The Android app needs your server for AI and sign-in checks. Play needs public web pages for privacy and account deletion.

1 Deploy to Vercel

Import the repo, framework Next.js, and add these environment variables (Production):

```
# Vercel → Project → Settings → Environment Variables
NEXT_PUBLIC_FIREBASE_API_KEY=...      NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=...
NEXT_PUBLIC_FIREBASE_PROJECT_ID=openstack-28bf7  NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=...
NEXT_PUBLIC_FIREBASE_APP_ID=...      NEXT_PUBLIC_GOOGLE_WEB_CLIENT_ID=...
ANTHROPIC_API_KEY=sk-ant-...          # server only, never NEXT_PUBLIC_
AI_MODEL=claude-haiku-4-5             AI_DAILY_LIMIT=20
```

2 Connect the domain

Vercel → Domains → add `stackforge.in` and follow the DNS steps. HTTPS is automatic.

3 Firebase for production

Firebase console → Authentication → Settings → **Authorized domains**: add `stackforge.in` and your Vercel URL. Then push the security rules:

```
firebase use openstack-28bf7
firebase deploy --only firestore:rules,storage # Storage only if you're on the Blaze plan
```

4 Public pages

Add three simple pages to the site:

`/privacy` : what's stored on the phone, what's synced (email, notes, highlights, PDFs if uploaded), that AI text goes to Anthropic for processing, and how to delete it.

`/delete-account` : steps to delete in the app, plus an email option for people who've uninstalled it.

`/` : a landing page with the logo, one line ("Stack makes you remember what you read"), screenshots and a "Get it on Google Play" button.

5 Smoke test

From a phone on mobile data (not your Wi-Fi): open the site, sign in with Google, save a note, run one AI summary and check that the note appears on the web. If any of these fail, fix it before Stage 4.



THE RELEASE BUILD

Play takes an **Android App Bundle (.aab)** signed with your **upload key**. Google then re-signs it with the app signing key it keeps (Play App Signing).

1 Create your upload key (once)

Keep it **outside** the repo.

```
keytool -genkeypair -v -keystore ~/keys/stackforge-upload.jks \
  -alias stack -keyalg RSA -keysize 2048 -validity 10000
# back up the .jks file and its passwords in 2 places (e.g. a password manager + a USB drive)
```

2 Tell Gradle about it

Create `android/keystore.properties`, add it to `.gitignore`, and read it in `android/app/build.gradle`:

```
# android/keystore.properties (never commit)
storeFile=/home/aman-kumar-chhari/keys/stackforge-upload.jks
storePassword=... keyAlias=stack keyPassword=...

// android/app/build.gradle
def ks = new Properties(); ks.load(new FileInputStream(rootProject.file("keystore.properties")))
android { signingConfigs { release { storeFile file(ks.storeFile); storePassword ks.storePassword
                                keyAlias ks.keyAlias; keyPassword ks.keyPassword } }
    buildTypes { release { signingConfig signingConfigs.release } } }
```

3 Build the bundle

Copy `build-apk.sh` to `build-aab.sh` and change the last steps:

```
STACK_TARGET=apk STACK_STORE=play npx next build
node scripts/android-setup.mjs && npx cap sync android
(cd android && ./gradlew bundleRelease)
# → android/app/build/outputs/bundle/release/app-release.aab
```

4 Fix Google sign-in for the store build

After the first upload, go to Play Console → **Test and release** → **App integrity** and copy the **app signing key SHA-1** and **upload key SHA-1**. Add both in Firebase → Project settings → your Android app → **Add fingerprint**. Without this, Google sign-in fails on Play installs even though it works on your debug APK.

5 Test the release build on a real phone

Install it through the internal testing track (next stage) and repeat the smoke test: sign-in, save, AI, sync, music, notifications, dark mode.



PLAY CONSOLE SETUP

Create app → name "**Stack: Read, Keep, Remember**" · App · Free · default language English (India). Then clear every item under **Policy** → **App content**.

FORM	WHAT TO ANSWER FOR STACK
Privacy policy	https://stackforge.in/privacy
App access	"All features work without login." If a feature needs sign-in, give a test Google account's instructions
Ads	No ads
Content rating	Fill the IARC questionnaire honestly. It's a reading and notes tool, but user content and web articles can include anything
Target audience	18+ (or 13+). Don't select under-13, or the stricter Families rules apply
Data safety	Collected: email and name (account), user content (notes, highlights, files if synced), app activity if you add analytics. Encrypted in transit: yes. Users can request deletion: yes (link /delete-account). Shared: AI text is processed by Anthropic on your behalf
Account deletion	URL https://stackforge.in/delete-account
Foreground service	Declare media playback (music during Focus): give a short description and a short screen recording of background play with its notification
App category	Education or Productivity . Not "News & Magazines", which brings extra news-publisher rules

STORE LISTING ASSETS

ASSET	SPEC	TIP
App icon	512 × 512 PNG	The block logo on paper colour
Feature graphic	1024 × 500	Reuse Stack-Brand/banner-wide
Phone screenshots	2–8, 9:16	Lead with Save → Highlight → Review → Quiz
Tablet screenshots	7" and 10"	You have tablet layouts, so use them
Short description	≤ 80 characters	"Save articles & PDFs, highlight, and actually remember what you read."
Full description	≤ 4,000 characters	Problem → 4 steps → features → free vs Plus → privacy



12 TESTERS × 14 DAYS

New personal developer accounts must run a **closed test** with at least **12 testers opted in continuously for 14 days** before Play lets you publish to everyone. Treat it as your first real user research, not a chore.

1

Internal testing first (day 0)

Test and release → Testing → **Internal testing** → upload the `.aab` → add yourself and 2–3 friends. It's live in minutes, so use it to catch install and sign-in problems.

2

Start the closed test (day 1)

Testing → **Closed testing** → create a track → add the tester emails (a Google Group is easiest) → upload the same build → send for review. The first review can take a few days.

3

Get testers opted in

Send each tester the opt-in link from the track page. Invite **15–20**, because anyone who opts out resets their own 14-day count. Check the tester count in the Console every few days.

4

Give them something to do

Day 1: "Save 3 articles or PDFs and highlight one line." Day 3: "Do the morning review." Day 7: a 5-question Google Form (What did you use? What confused you? Would you miss it? Would you pay ₹99? What's missing?).

5

Ship 1–2 updates during the test

Fix the top complaints and upload new builds to the same track (bump `versionCode`). Google looks at whether you actually tested and improved the app.

6

Apply for production (day 15+)

Dashboard → **Apply for production**. Google asks how you recruited testers, what feedback you got and what you changed, so answer from your notes and form results. A review usually takes about a week.

WHERE TO FIND TESTERS

College and alumni WhatsApp groups · dev.to / Reddit r/androidapps "looking for testers" posts · Discord servers you're in · friends who read a lot · people who used Pocket. Offer free Stack Plus for a year as thanks.



LAUNCH DAY

Once production access is approved, launch in a controlled way and make noise in the same week.

1

Staged rollout

Production → create release → rollout **20%**. Watch Android vitals (crashes, ANRs) for 2 days → **50%** → **100%**. You can halt a rollout if something breaks.

2

Make the GitHub repo public

Before flipping it: run a secret scan again, add a **LICENSE** (**AGPL-3.0** protects the open core from closed clones; MIT is simpler if you want maximum reuse), a `CONTRIBUTING.md`, and a README that links to the Play Store and stackforge.in.

3

Update the landing page

Add the Play badge, 3 screenshots, a 30-second screen recording and an email signup for "Stack Plus is coming".

4

Post the launch story

The same week, post on: LinkedIn (build story + video), dev.to ("I built a local-first reading app with Next.js + Capacitor"), Reddit (r/androidapps, r/IndiaTech, r/productivity; follow each community's self-promotion rules), X, college groups, and Product Hunt on a Tuesday–Thursday.

5

Ask your testers for reviews

Your 12–20 testers can now leave public reviews. Early 4–5★ reviews matter a lot for store ranking.

SAMPLE LAUNCH POST

I read a lot of articles and PDFs, and I forgot almost all of it. So I built **Stack**.

Save anything from Chrome, WhatsApp or Telegram → highlight a line → Stack brings it back the next morning as a quick review and quiz.

Free, open source, and it works offline without an account. Made in India by **StackForge Labs**.

► [PLAY STORE LINK](#) • ★ [GITHUB.COM/CHHARI07/STACKFORGE](https://github.com/CHHARI07/STACKFORGE) • [30-SEC DEMO VIDEO](#)



THE FIRST 30 DAYS

Launch is the start of learning. Keep a steady weekly rhythm and only fix what users actually hit.

EVERY DAY · 15 MIN

- ☐ Reply to every Play review and email (even 1★). Thank people and say what you'll fix
- ☐ Glance at crashes/ANRs in Android vitals and the Anthropic usage page

EVERY SUNDAY · 1 HOUR

- ☐ Write down: installs, day-1 / day-7 / day-30 active, reviews done per user, AI cost
- ☐ Pick the **top 3** problems from reviews, the form and analytics. Those are next week's work

EVERY WEEK · ONE UPDATE

- ☐ Small, frequent releases (bump `versionCode`, short "What's new")
- ☐ One public post a week: a feature, a lesson learned or a number you hit

DAY 30 · DECIDE

- ☐ Day-30 retention $\geq 25\%$: keep going and start Stack Plus (Play Billing) in month 2
- ☐ **10-25%**: interview 10 users; change the core loop, not the marketing
- ☐ **< 10%**: rethink who Stack is for before adding anything new

WHEN THINGS GO WRONG

PROBLEM	WHAT TO DO
Google sign-in fails on Play installs	The Play app signing SHA-1 is missing in Firebase (Stage 4, step 4)
Play rejects the app	Read the email carefully, fix only that item, reply through the Policy status page. Don't resubmit unchanged
AI bill jumps	Lower <code>AI_DAILY_LIMIT</code> ; check for loops; the Anthropic spend cap stops it anyway
A bad release is out	Halt the staged rollout, fix, upload a higher <code>versionCode</code>
Lost the upload key	Play Console → App integrity → request an upload key reset. The app signing key is safe with Google



THE CHECKLIST

Every step from this guide on one page.

STAGE 1 · ACCOUNTS

- ☐ StackForge trademark search (IP India, classes 9 + 42)
- ☐ Domains bought: stackforge.in + one global
- ☐ Play Console account created and verified (\$25)
- ☐ Support email on your domain
- ☐ Anthropic key + monthly spend limit
- ☐ Vercel linked to the GitHub repo
- ☐ 15–20 tester emails collected

STAGE 2 · APP

- ☐ Play build without `MANAGE_EXTERNAL_STORAGE`
- ☐ Spotify hidden or "bring your own ID"
- ☐ News: summary + link; full reader for saved items
- ☐ AI on Haiku + daily limit
- ☐ AI URL = https domain; cleartext only in debug
- ☐ Delete account inside the app
- ☐ Analytics for day-1/7/30 active
- ☐ `versionCode` / `versionName` set; changes committed

STAGE 3 · SERVER

- ☐ Vercel env vars set (key is server-only)
- ☐ Domain connected, HTTPS on
- ☐ Firebase authorized domains updated
- ☐ Firestore / Storage rules deployed
- ☐ `/privacy`, `/delete-account` and landing page live
- ☐ Smoke test on mobile data passed

STAGE 4 · RELEASE BUILD

- ☐ Upload keystore made and backed up twice
- ☐ `keystore.properties` gitignored
- ☐ `bundleRelease` produces `app-release.aab`
- ☐ Play SHA-1s added to Firebase

STAGE 5 · PLAY CONSOLE

- ☐ Privacy, app access, ads, rating, audience
- ☐ Data safety + account deletion URL
- ☐ Foreground service (media) declared
- ☐ Category: Education / Productivity
- ☐ Icon, feature graphic, screenshots, descriptions

STAGE 6 · TESTING

- ☐ Internal test installed and working
- ☐ Closed test live with 15–20 invited
- ☐ 12+ opted in for 14 straight days
- ☐ Feedback form sent; 1–2 updates shipped
- ☐ Applied for production

STAGE 7 · LAUNCH

- ☐ Staged rollout 20% → 50% → 100%
- ☐ Repo public with LICENSE + README
- ☐ Launch posts published
- ☐ Testers asked for reviews
- ☐ Sunday numbers routine started



RULES & NOTES

NEVER

- Commit `.env.local`, `google-services.json`, the `.jks` or `keystore.properties`
- Put `ANTHROPIC_API_KEY` in a `NEXT_PUBLIC_` variable (it would ship inside the app)
- Upload a debug-signed build to Play
- Change the app ID `com.chhari.stack` after publishing: it would become a new app
- Ask testers for fake reviews or pay for installs

ALWAYS

- Build releases from a clean, committed state and tag them (`git tag v1.0.0`)
- Keep a `CHANGELOG.md`: it becomes the "What's new" text
- Test every release on a real phone, on mobile data
- Keep the sideload APK and the Play build on the same version numbers
- Answer users within a day in the first month

THINGS TO WATCH

TOPIC

NOTE

Organisation account later

Once StackForge Labs is registered (LLP / Pvt Ltd) and has a D-U-N-S number, you can move the app to an organisation account. New apps from organisation accounts skip the 12-tester rule

Sideloaded APK

Google is rolling out developer verification for apps installed outside Play, starting with a few countries in 2026 and going wider in 2027. Plan to register as a verified developer so your APK keeps installing

DPDP Act

Full rules apply from **13 May 2027**: clear consent, data export/delete, breach reporting. The privacy page and delete-account flow cover the basics

Target SDK

Play raises the minimum target level every year (usually by 31 August). Update `targetSdkVersion` each summer

Stack Plus billing

Subscriptions inside the Android app must use **Google Play Billing** (15% fee on subscriptions). On the web you can use Razorpay

RELATED

StackForge_Stack_Plan.pdf: the business plan and market research · **ROADMAP.md**: feature roadmap · **firebase/README.md**: Firebase setup.

Play rules change often. Before each stage, check the Play Console Help pages for the latest version of each requirement.